Example

Create the following program:

Open the ”Serial Monitor” in order to se the output

TRY IT OUT!

```
int z;int a;int b;
void setup()
{
     Serial.begin(9600);
}
void loop()
{
     a = random(100);
     b = random(100);
     z = calculate(a,b); //Adding 2 Numbers

     //Write Values to Serial Monitor
     Serial.print(a);
     Serial.print(" + ");
     Serial.print(b);
     Serial.print(" = ");
     Serial.println(z);

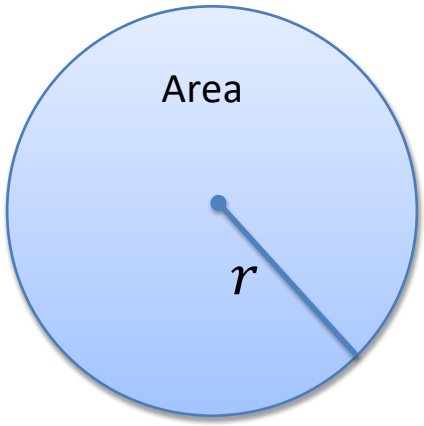
     delay(1000);
}
float calculate(int x, int y)
{
     return (x + y);
}
```

Creating Functions

TRY IT OUT!

Create a function that calculates the area of a circle with a given radius.

Write the result to the Serial Monitor.



Solution

TRY IT OUT!

```
void setup()
{
    float area;

    Serial.begin(9600);
    // calculate the area of a circle with radius of 9.2
    float r=9,2;
    area = CircleArea(r);
    Serial.print("Area of circle is: ");
    // print area to 4 decimal places
    Serial.println(area, 4);
}

void loop()
{

}

// calculate the area of a circle
float CircleArea(float radius)
{
    float result;
    const float pi = 3.14;

    result = pi * radius * radius;

    return result;
}
```

For Loop

In this program we use a For Loop to find the Sum of 100 Random Numbers.

Then we find the Average.

The Sum and Average should be written to the Serial Monitor.

TRY IT OUT!

```
int x; int sum = 0; float gjennomsnitt = 0;
void setup()
{
    Serial.begin(9600);
}

void loop()
{
    sum = 0;
    for (int i = 0; i<100; i++)
    {
        x = random(100);
        sum = sum + x;
    }

    average = sum / 100;
    Serial.print(" Sum = ");
    Serial.print(sum);
    Serial.print(" ,
    Average = ");
    Serial.println(average);
    delay(1000);
}
```

Arrays

Here we shall use arrays in the Arduino program

Create this program from scratch and open the Serial Monitor to see the result.

TRY IT OUT!

```
const int arraysize = 100;
int x;
int sum = 0;
float average = 0;
int myarray[arraysize];

void setup()
{
	Serial.begin(9600);
}

void loop()
{
	sum = 0;
	for (int i = 0; i < arraysize; i++)
	{
		x = random(200);
		myarray[i] = x;
	}
	sum = calculateSum(myarray);
	average = sum / 100;
	Serial.print(" Sum = ");
	Serial.print(sum);
	Serial.print(" , Average = ");
	Serial.println(average);
	delay(1000);
}

int calculateSum (int sumarray[])
{
	for (int i = 0; i < arraysize; i++)
	{
		sum = sum + sumarray[i];
	}
	return sum;
}
```

Arduino Programming



Here you will find complete overview of the Arduino programming language:

https://www.arduino.cc/en/Reference/HomePage

Play and Explore

|Programming with Arduino|

https://www.halvorsen.blog



Arduino Examples

|Hans-Petter Halvorsen |

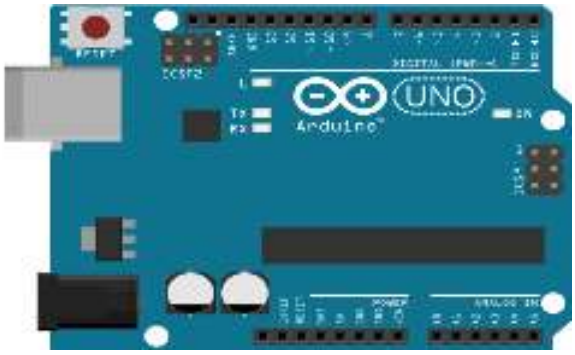
Arduino Examples

On the next pages you will find some Examples/Tasks that you should do step by step as stated in the text.

When you have finished a specific Example/Task, it is recommended that you “Play and Explore”, i.e., make small changes in the program, etc.

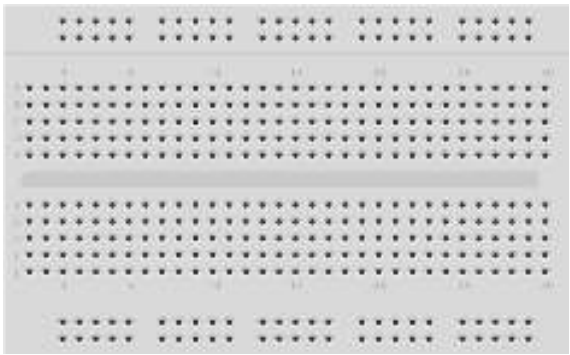
We need the following

Arduino UNO



+

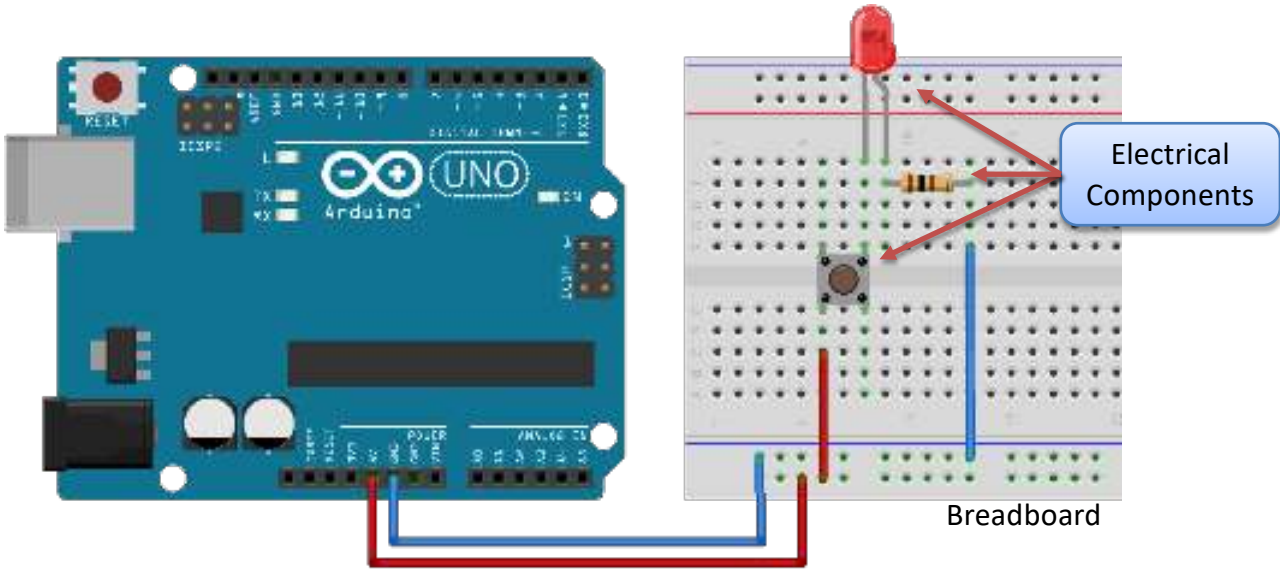
Breadboard



+ LEDs, Resistors, Wires, etc.

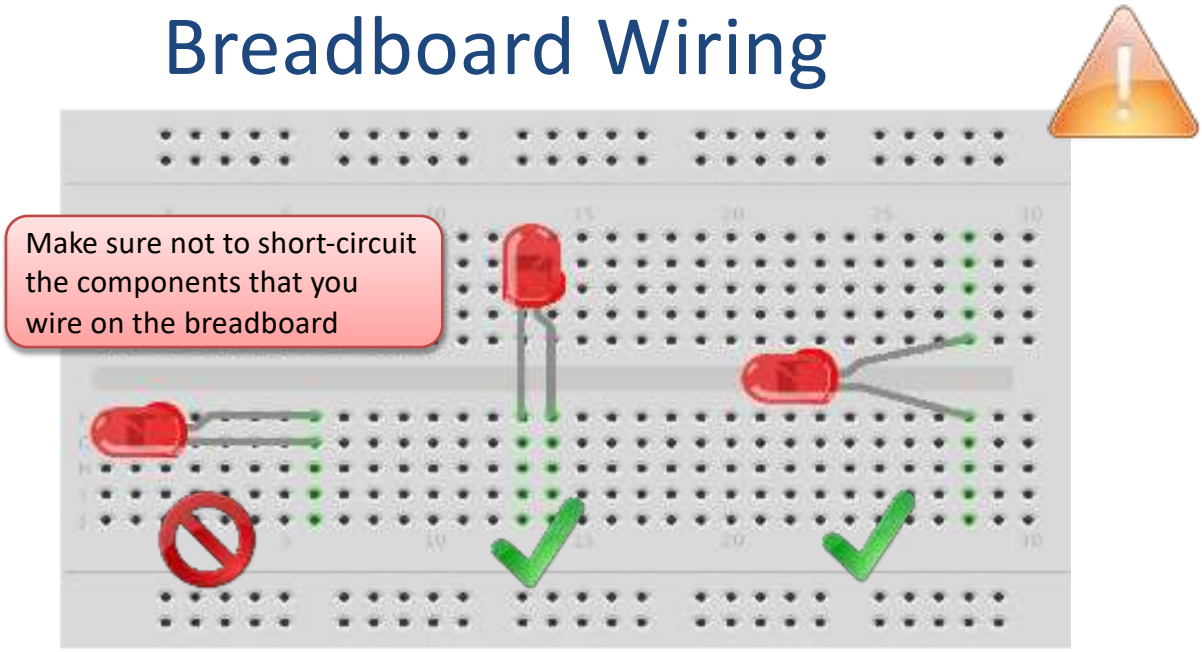
The Breadboard is used to connect components and electrical circuits

Breadboard Example



Wires used to connect the electrical components together to a closed circuit

Breadboard Wiring



The Breadboard is used to connect components and electrical circuits

Resistors

TRY IT OUT!



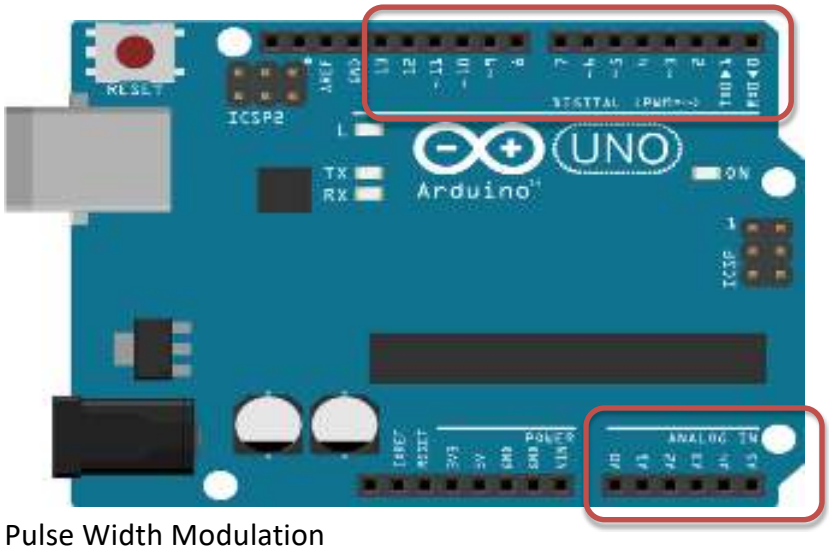
What is the values for your resistors?

- Use the Color Codes to figure it out
- Or use also a Multimeter to see if you get the same results.
- Or use a Resistor Calculator

http://www.allaboutcircuits.com/tools/resistor-color-code-calculator/

Inputs and Outputs (Analog and Digital)

Digital Inputs and Digital Outputs



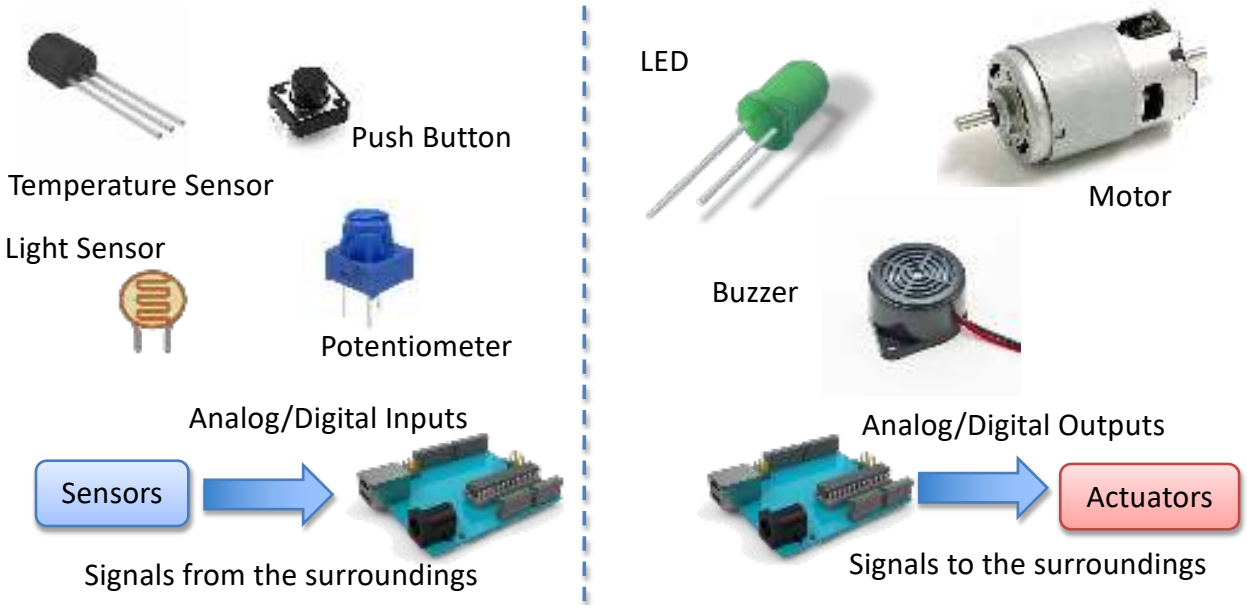
You can choose from the code if they are to be inputs or outputs

Those marked with ~ can also be used as "Analog Outputs", so-called PWM outputs

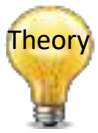
Analog Inputs

PWM - Pulse Width Modulation

Sensors and Actuators



Sensors and Actuators

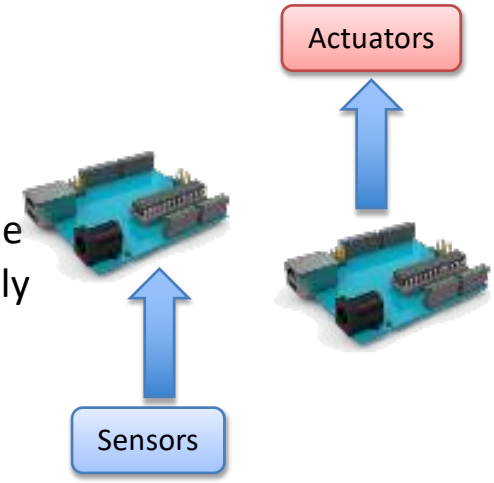


- A **Sensor** is a converter that measures a physical size and converts it to a signal that can be read by an instrument, data acquisition device, or an Arduino.
  Examples: temperature sensor, pressure sensor, etc.
- An **Actuator** is a kind of motor that moves or controls a mechanism or system. It is powered by an energy source, typical electric current, hydraulic fluid pressure, or air pressure, and converts this energy into motion.
  Examples: Engine, Pump, Valve, etc.

http://en.wikipedia.org/wiki/Sensor http://en.wikipedia.org/wiki/Actuator

Sensors and Actuators

- The sensors and actuators can be either digital or analog.
- Some sensors and actuators have been made for Arduino, while others need to be connected in some circuit to work properly with Arduino.
- Many of these come with ready-made libraries for Arduino, so they are easy to use.



Examples

1. Electrical Circuit Example
2. Blinking LED Example
3. Switch Example
4. Potentiometer Example
5. Temperature Example
6. Light Sensor Example
7. Thermistor Example

Play and Explore

|Programming with Arduino|

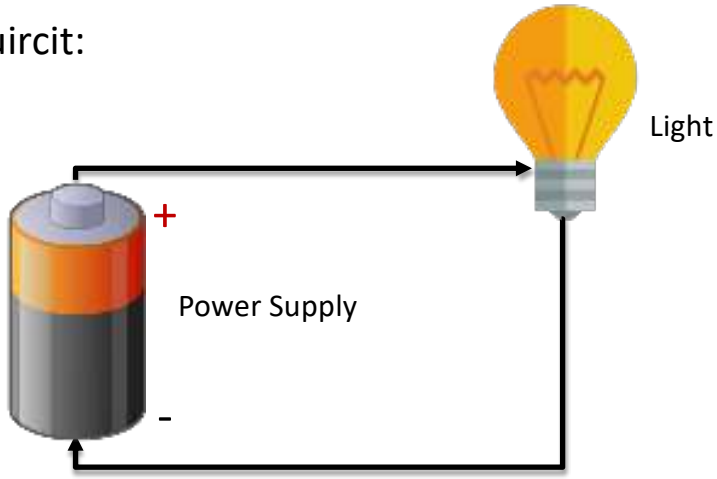
Example 1

TRY IT OUT!

Electrical Circuits

Electrical Cuicits

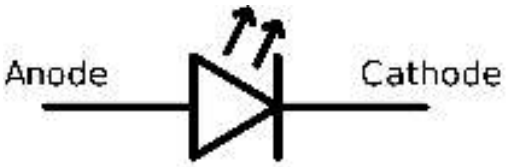
Lets create the following cuircit:

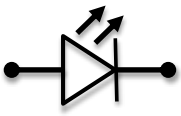


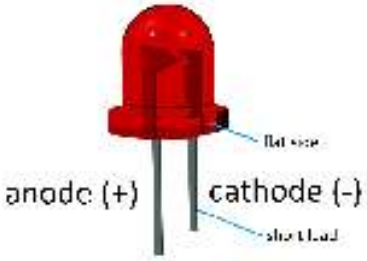
Instead of using a Battery we will use the Arduino board as a Power Supply (5V)

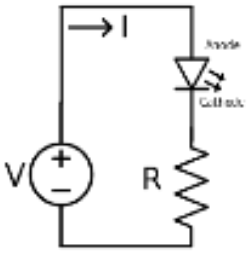
Example 1

Light-Emitting Diode - LED





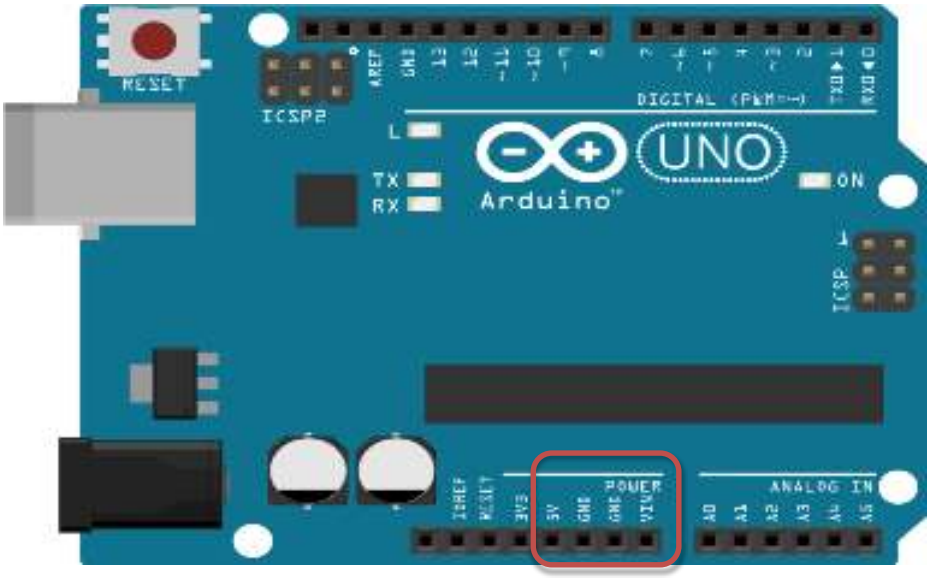






[Wikipedia]

Introduction



We will use the POWER ports ”5V” and ”GND” on the Arduino board

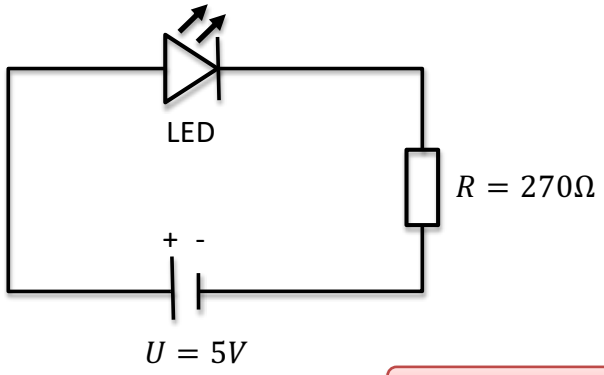
GND = Ground

Example 1

Electrical Cuircuit

TRY IT OUT!

Make the following circuit using the Arduino board and a Breadboard:



Equipment:
- Breadboard
- LED
- Resistor
- Wires
- Multi-meter

Note! No Arduino Program is needed in this example

Example 1

Why do you need a Resistor?



If the current becomes to large, the LED will be destroyed. To prevent this to happen, we will use a Resistor to limit the amount of current in the cuircuit.

What should be the size of the Resistor?

A LED typically need a current like 20mA (can be found in the LED Datasheet).
We use Ohm's Law:

$$U = RI$$

Arduino gives U=5V and I=20mA. We then get:

$$R = \frac{U}{I}$$

The Resistor needed will be $R = \frac{5V}{0.02A} = 250\Omega$. Resistors with R=$250\Omega$ is not so common, so we can use the closest Resistors we have, e.g., 270Ω

Example 1

Find the Resistor Size

Find the correct Resistor by using the Color Codes or a Mult-meter:

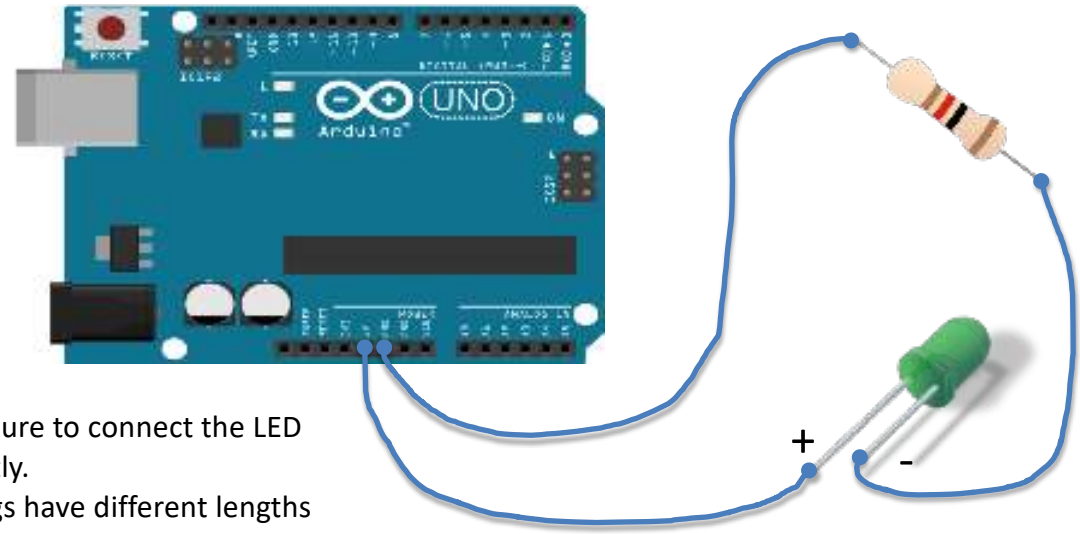


You may also use a "Resistor Calculator" which you find online:

http://www.allaboutcircuits.com/tools/resistor-color-code-calculator/

Example 1

Wiring

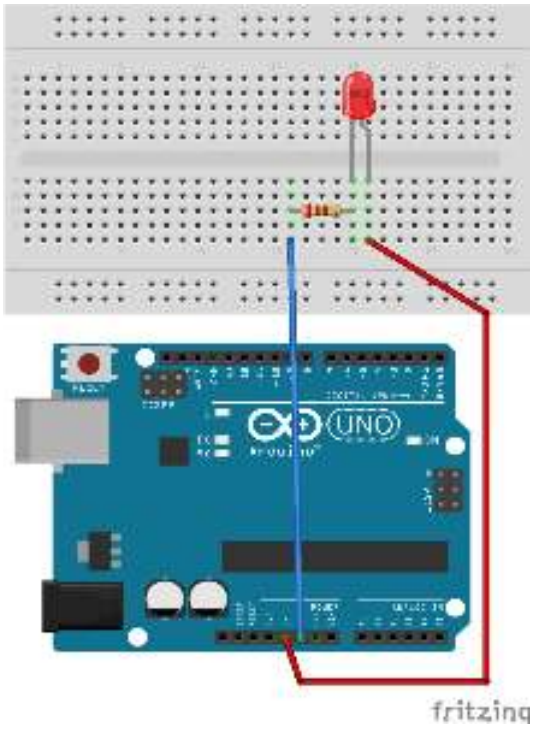


Make sure to connect the LED correctly.
The legs have different lengths

Example 1

Use a Breadboard – see next page

Wiring



TRY IT OUT!

Make sure to connect the LED correctly.
The legs have different lengths

Note! Use a Resistor $R = 270\Omega$ to protect the LED

Example 1

Play and Explore

|Programming with Arduino|



|Programming with Arduino|

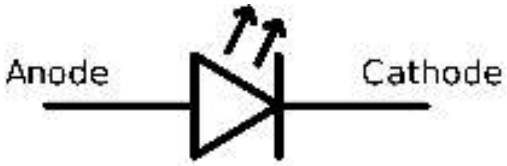
Example 2

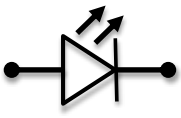
TRY IT OUT!

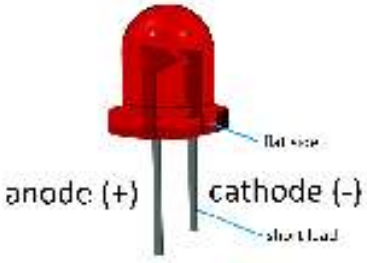
Blinking LED

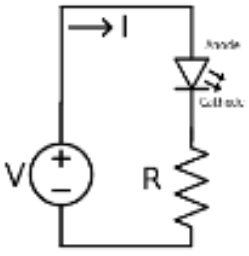


Light-emitting diode - LED











[Wikipedia]

Introduction

We will make a program that makes the LED start blinking.

How-To Do it:

1. Wire the cuircuit and components
2. Make the Arduino program

Example 2

Equipment

- Arduino UNO
- Breadboard
- LED
- Resistor, $R = 270\Omega$
- Wires (Jumper Wires)







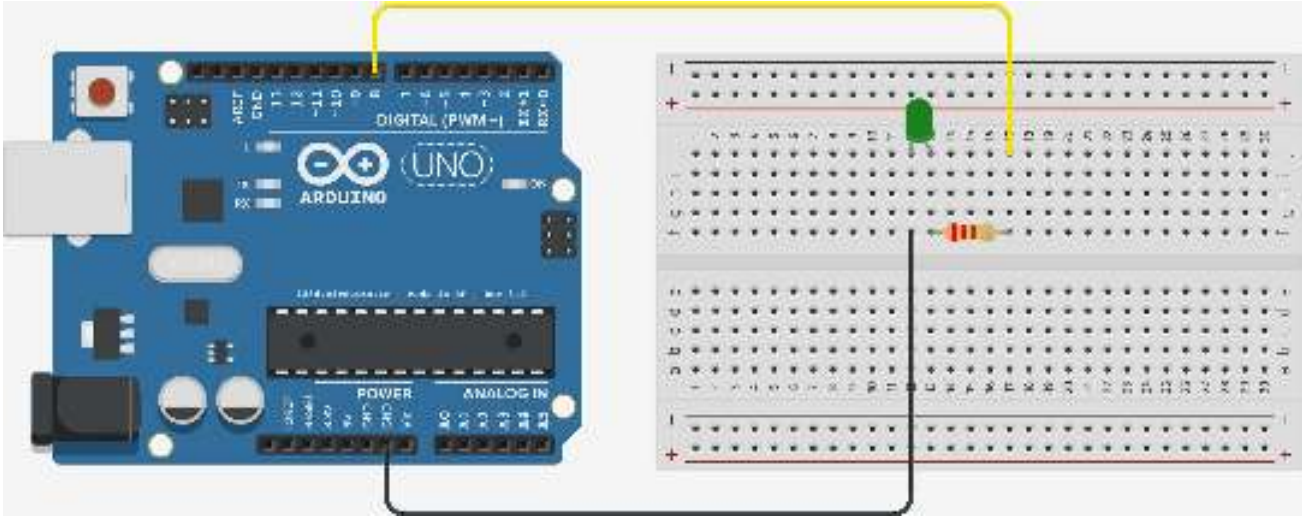




Example 2

Wiring

TRY IT OUT!



Example 2

Programming

Program Structure

```
//Globale variable
...

void setup()
{
  //Initialization
}

void loop()
{
  //Main Program
}
```

You need to use the following:

Which Pin (0, 1, 3, ...) are you using?

A Digital Pin can either be an INPUT or an OUTPUT. Since we shall use it to turn-on a LED, ww set it to OUTPUT.

```
pinMode(pin, mode);
```

```
digitalWrite(pin, value);
```

Turn-on LED

Turn-off LED

A Digital PIn can have 2 values, either HIGH or LOW

```
delay(ms);
```

The delay() fuction make a small pause in milliseconds (ms), e.g,. delay(1000) pause the program for 1 second

Example 2

Arduino Language Reference: https://www.arduino.cc/en/Reference

Arduino Program

Example 2

TRY IT OUT!

```
void setup()
{
   pinMode(8, OUTPUT);
}

void loop()
{

   digitalWrite(8, HIGH);    // Turn on the LED
   delay(1000);              // Wait for one second
   digitalWrite(8, LOW);     // Turn off the LED
   delay(1000);              // Wait for one second
}
```

Arduino Program

Example 2

TRY IT OUT!

```
int ledPin = 8;

void setup()
{
   pinMode(ledPin, OUTPUT);
}

void loop()
{

   digitalWrite(ledPin, HIGH);   // Turn on the LED
   delay(1000);                  // Wait for one second
   digitalWrite(ledPin, LOW);    // Turn off the LED
   delay(1000);                  // Wait for one second

}
```

En ørliten forbedring. Vi bruker en variabel til å definere pinne-nummeret

Play and Explore

|Programming with Arduino|



|Programming with Arduino|

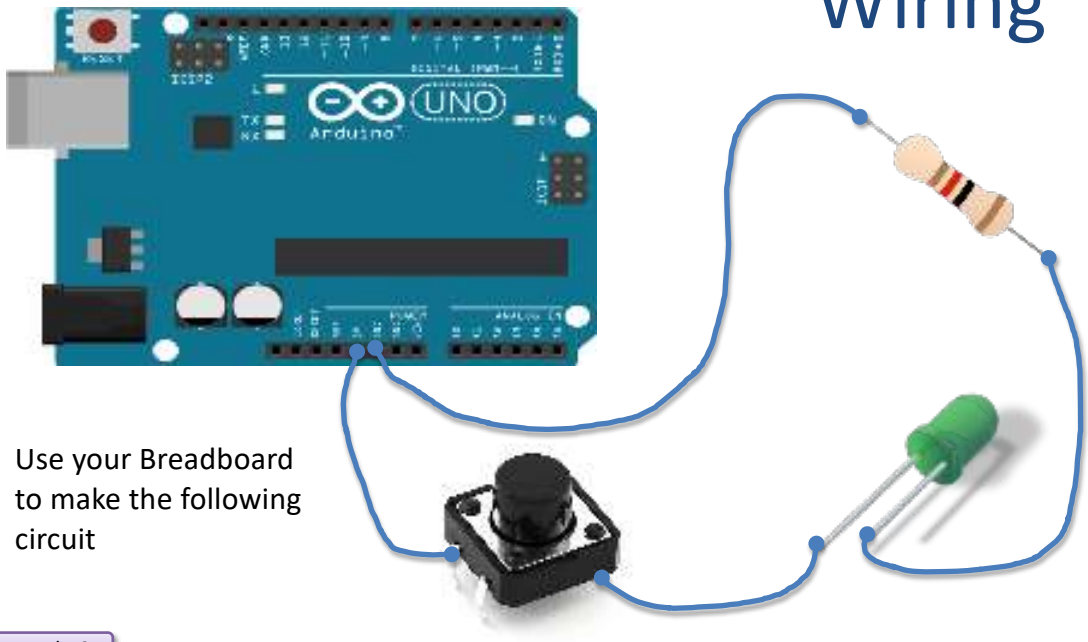
Example 3

TRY IT OUT!

Switch

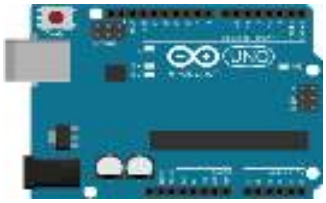


Wiring



Use your Breadboard to make the following circuit

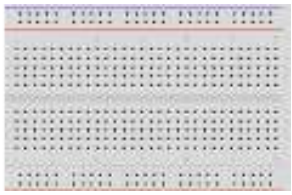
Equipment



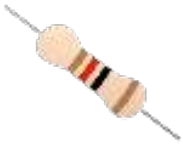
- Arduino
- Breadboard
- LED
- Switch
- Resistor, $R = 270\ \Omega$
- Some Wires







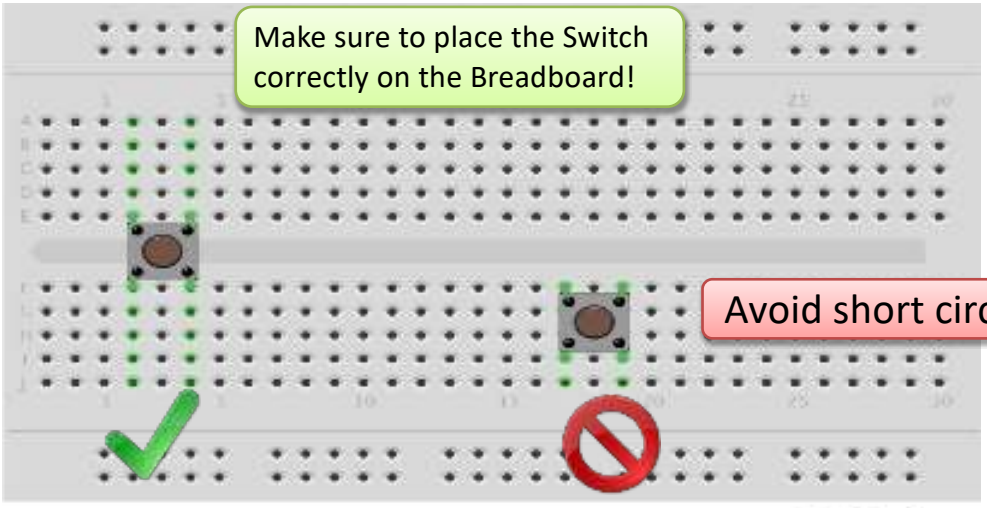




Example 3

Breadboard



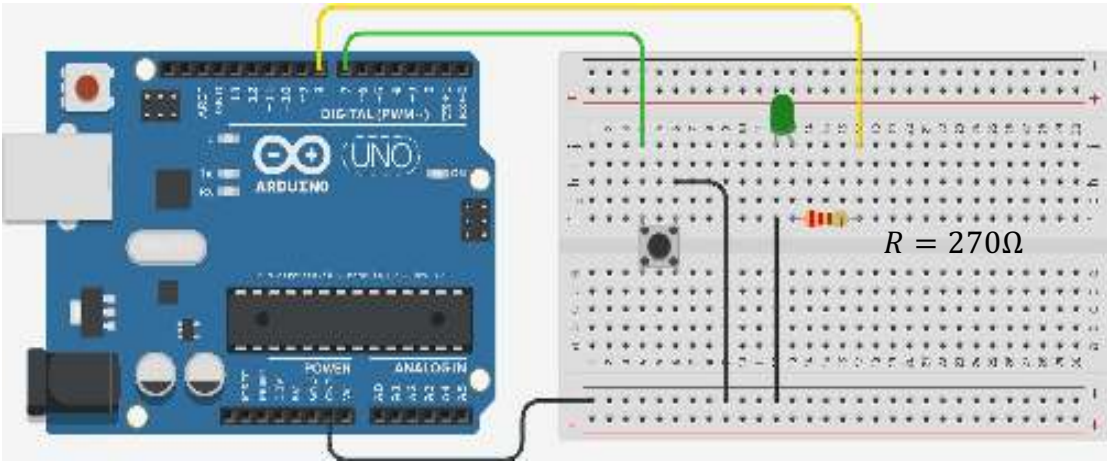


Avoid short circuit

Example 3

TRY IT OUT!

Wiring



Note! In this configuration, we use an internal "pull-up" resistor to prevent "short-circuiting".

Example 3

Set pinMode like this: `pinMode(pin, INPUT_PULLUP);`

Arduino Program

TRY IT OUT!

```
const int buttonPin = 7;
const int ledPin =  8;

int buttonState = 0;

void setup()
{
    pinMode(ledPin, OUTPUT);

    pinMode(buttonPin, INPUT_PULLUP);
}

void loop()
{
  buttonState = digitalRead(buttonPin);


  if (buttonState == HIGH)
  {
        digitalWrite(ledPin, HIGH);
  }
  else
  {
        digitalWrite(ledPin, LOW);
  }

}
```

Example 3